

census_and_LODES_data.csv & ZillowHousingData.csv Data Management: **Data Merging, Cleaning, & Separating**

Zillow

1. Zillow Merge (Right merge with crosswalk on zip_code and RegionName - Zip code for Zillow) to get all Zillow data for the Bay Area (Covered by the crosswalk range)
2. Create city indicator variable for (SF, San Jose, Oakland)
3. Take January beginning and ending averages across the 5-year zones (2006 & 2010, 2011 & 2015, 2016 & 2020)
 - a. Leave 2021-2026 data as is for now

Census & LODES

4. Drop rows with missing data for 2010, 2015, or 2020 (census or LODES)
 - a. Dropped about 400 rows
5. Separate 2022 LODES data
 - a. Create a new dataframe and CSV, then later add this to the Census data for 2022, 2023, 2024 (test dataset)
6. Create CSV files for training data, validating data, and LODES 2022 testing data (To be added to Census later)

Final Merges

1. Training Data: Merge the 2010 & 2015 Zillow home value measurements with the Census & LODES data
2. Validation Data: Merge the 2020 Zillow home value measurements with the Census & LODES data
3. Create a descriptor dataframe for general reference if we need

Adding Gentrification Measures

1. Find City-specific median income and create an additional column (city_median_income 2010/2015/2020)
2. Create gentrification criteria columns – change in: average income, home values, rent, education percentage
 - a. Added to each dataset (training and validation)
3. Added gentrification indication to each tract based on the number of criteria met
 - a. $\frac{3}{4}$ is a positive indicator for gentrification

To Do:

- Add testing census data (2022-2024), and run the same merges on that data
 - Calculate the three-year averages to make it look like the 5-year averages from training and validation steps
 - Merge with 2022 LODES data
 - Merge with Zillow data and gather January columns
 - Add gentrification measures
- Need to separate the 2010 and 2015 data into separate rows in the training data to prevent data leakage

- Where are we using Zillow Data? - Double Check

Limitations:

- The 2024 ACS data is a 5-year average, meaning it overlaps with the validation data for the 2020 data
 - Since we are dealing with averages, this data is not as crucial and isn't directly represented in both
 - Good to show the COVID data in both validation and testing steps